Why Regularization Works

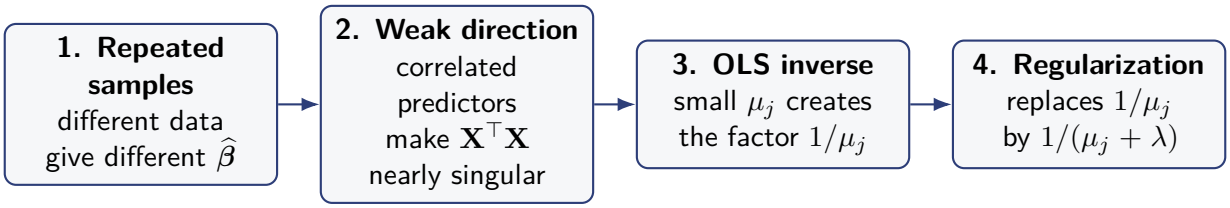
Ridge, Lasso, and the weak directions of linear regression

Machine Learning 2026 ▪ Lecture 7

Typed class notes from the 1 September board derivation

The question that organizes the lecture

Ordinary least squares can fit the training data well and still produce coefficients that change dramatically when the training sample changes slightly. **Why does this happen, and how does a penalty make the model more stable?**



Notation used throughout

$\mathbf{X} \in \mathbb{R}^{n\times d}$	design matrix: n observations, d predictors
$\mathbf{y} \in \mathbb{R}^n$	observed responses
$\boldsymbol{\beta} \in \mathbb{R}^d$	model coefficients; predictions are $\widehat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta}$
$\mathbf{q}_j$	eigen-direction of $\mathbf{X}^\top\mathbf{X}$
μ_j	eigenvalue of $\mathbf{X}^\top\mathbf{X}$ along $\mathbf{q}_j$
$\lambda \geq 0$	regularization strength, not an eigenvalue

One notation choice prevents several later confusions

The board work used both θ and β for coefficients. These notes use $\boldsymbol{\beta}$ only. We reserve μ_j for eigenvalues and λ for the penalty strength.

1. Why regularize?

The three motivations written at the start of the board work

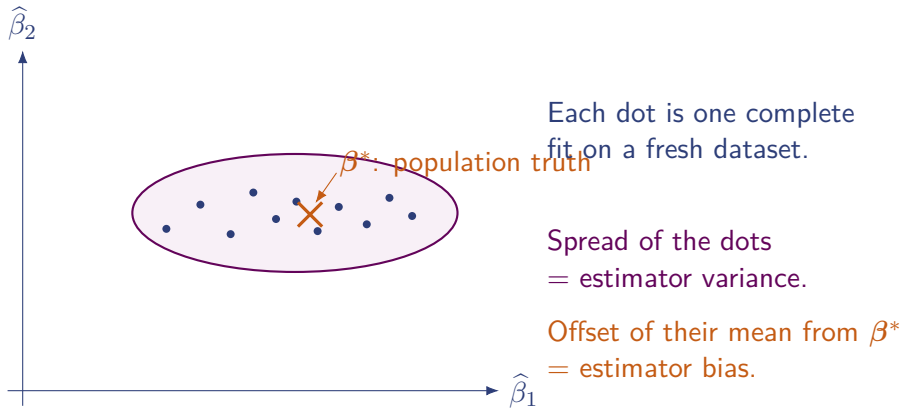
Bias–variance tradeoff
accept a little bias
to reduce instability

Feature selection
prefer a simpler set
of active coefficients

Penalization
encode a preference
for smaller coefficients

Suppose an unknown population coefficient vector is $\boldsymbol{\beta}^*$. A training dataset D produces an estimate $\widehat{\boldsymbol{\beta}}(D)$. If we repeatedly draw new training datasets from the same population, we generally obtain different estimates:

$$D^{(1)}, D^{(2)}, \ldots, D^{(R)} \quad \longrightarrow \quad \widehat{\boldsymbol{\beta}}^{(1)}, \widehat{\boldsymbol{\beta}}^{(2)}, \ldots, \widehat{\boldsymbol{\beta}}^{(R)}.$$



Bias and variance of a coefficient estimator

$$\mathrm{Bias}(\widehat{\boldsymbol{\beta}}) = \mathbb{E}_D[\widehat{\boldsymbol{\beta}}(D)] - \boldsymbol{\beta}^*, \qquad \mathrm{Var}(\widehat{\boldsymbol{\beta}}) = \mathbb{E}_D\left[\left\|\widehat{\boldsymbol{\beta}}(D) - \mathbb{E}_D[\widehat{\boldsymbol{\beta}}(D)]\right\|_2^2\right].$$

Bias asks where the cloud is centered. Variance asks how widely the fitted coefficients move across repeated samples.

The regularization bargain

We deliberately pull coefficients toward a simpler reference point, usually zero. This introduces some bias, but it can reduce variance enough to improve prediction on new data.

2. Begin with ordinary least squares

Derive the object that regularization will modify

The linear model predicts all n responses at once:

$$\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta}, \qquad \mathbf{r} = \mathbf{X}\boldsymbol{\beta} - \mathbf{y}.$$

Ordinary least squares chooses the coefficients that minimize squared residual length:

$$J_{\mathrm{OLS}}(\boldsymbol{\beta}) = \|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2 = (\mathbf{X}\boldsymbol{\beta} - \mathbf{y})^\top(\mathbf{X}\boldsymbol{\beta} - \mathbf{y}).$$

Step 1: expand the scalar objective

$$\begin{aligned} J_{\mathrm{OLS}}(\boldsymbol{\beta}) &= \boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{y} - \mathbf{y}^\top\mathbf{X}\boldsymbol{\beta} + \mathbf{y}^\top\mathbf{y} \\ &= \boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta} - 2\boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{y} + \mathbf{y}^\top\mathbf{y}. \end{aligned}$$

The two middle terms are equal because each is the same scalar written in transposed form.

Step 2: differentiate and set the gradient to zero

Because $\mathbf{X}^\top\mathbf{X}$ is symmetric,

$$\nabla_{\boldsymbol{\beta}} J_{\mathrm{OLS}} = 2\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta} - 2\mathbf{X}^\top\mathbf{y} = 2\mathbf{X}^\top(\mathbf{X}\boldsymbol{\beta} - \mathbf{y}).$$

At an optimum $\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}$,

$$\boxed{\mathbf{X}^\top\mathbf{X}\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} = \mathbf{X}^\top\mathbf{y}} \qquad \text{(the normal equation)}.$$

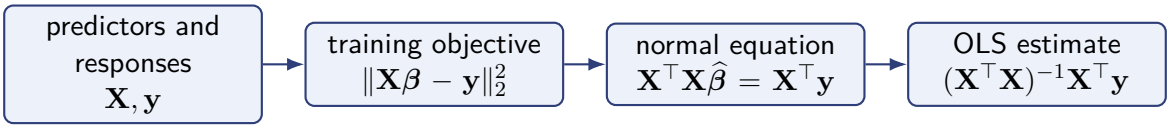
Step 3: solve only when the inverse exists

If the columns of $\mathbf{X}$ are linearly independent, $\mathbf{X}^\top\mathbf{X}$ is invertible and

$$\boxed{\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}.}$$

The matrix $\mathbf{X}^\top\mathbf{X} \in \mathbb{R}^{d\times d}$ is called the **Gram matrix**. Its entries are predictor inner products:

$$\mathbf{X}^\top\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top\mathbf{x}_1 & \cdots & \mathbf{x}_1^\top\mathbf{x}_d \\ \vdots & \ddots & \vdots \\ \mathbf{x}_d^\top\mathbf{x}_1 & \cdots & \mathbf{x}_d^\top\mathbf{x}_d \end{bmatrix}.$$



The inverse is exactly where instability can enter

The formula is not automatically reassuring. If $\mathbf{X}^\top\mathbf{X}$ has a very small eigenvalue, its inverse has a very large eigenvalue. The next two sections make this statement concrete.

3. Correlated predictors create a weak direction

The coefficients can move while predictions barely change

Consider two predictors whose columns are nearly identical:

$$\mathbf{x}_2 \approx \mathbf{x}_1.$$

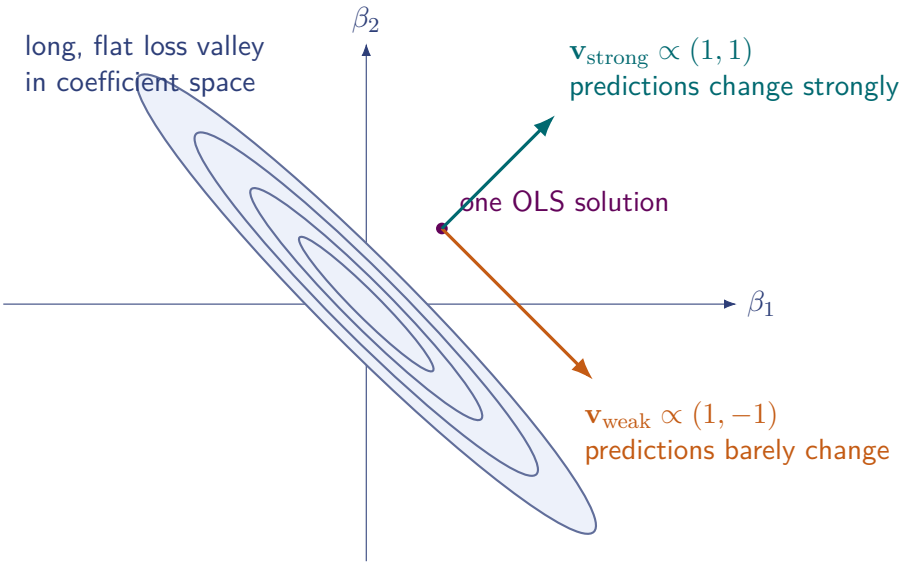
The prediction is

$$\hat{\mathbf{y}} = \beta_1 \mathbf{x}_1 + \beta_2 \mathbf{x}_2 \approx (\beta_1 + \beta_2)\mathbf{x}_1.$$

The data can therefore identify the *sum* $\beta_1 + \beta_2$ much more reliably than the two coefficients separately.

coefficient pair	sum	approximate prediction
$(4, 4)$	8	$8\mathbf{x}_1$
$(10, -2)$	8	$8\mathbf{x}_1$
$(10000, -9992)$	8	$8\mathbf{x}_1$

All three coefficient vectors can make nearly the same predictions. The extreme third pair is not a typographical error: it exposes what collinearity permits.



Define the unit direction

$$\mathbf{v}_{\text{weak}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Moving coefficients by $t\mathbf{v}_{\text{weak}}$ changes predictions by

$$\mathbf{X}(\boldsymbol{\beta} + t\mathbf{v}_{\text{weak}}) - \mathbf{X}\boldsymbol{\beta} = t\mathbf{X}\mathbf{v}_{\text{weak}} = \frac{t}{\sqrt{2}}(\mathbf{x}_1 - \mathbf{x}_2) \approx \mathbf{0}.$$

Physical meaning of a weak direction

The coefficient vector can move a long distance along $\mathbf{v}_{\text{weak}}$, while the fitted responses and training loss move very little. A new sample or small response perturbation can therefore push OLS far along the valley.

Check your understanding

If $\mathbf{x}_1 = \mathbf{x}_2$ exactly, verify that $\mathbf{X}[1, -1]^\top = \mathbf{0}$. Then explain why OLS cannot uniquely determine β_1 and β_2.

4. The spectral view exposes the instability

Rotate to the natural directions of the Gram matrix

The Gram matrix is symmetric and positive semidefinite:

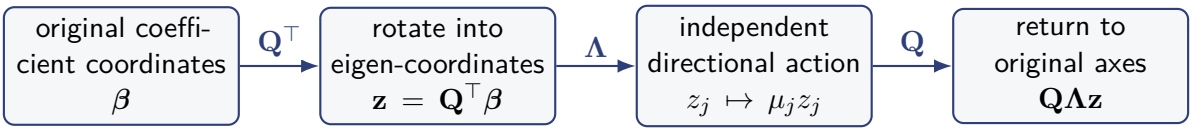
$$(\mathbf{X}^\top\mathbf{X})^\top = \mathbf{X}^\top\mathbf{X}, \qquad \mathbf{v}^\top\mathbf{X}^\top\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|_2^2 \geq 0.$$

The spectral theorem therefore gives

$$\boxed{\mathbf{X}^\top\mathbf{X} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top}, \qquad \mathbf{\Lambda} = \operatorname{diag}(\mu_1, \ldots, \mu_d), \qquad \mathbf{Q}^\top\mathbf{Q} = \mathbf{I}.$$

The columns $\mathbf{q}_j$ of $\mathbf{Q}$ are orthonormal eigen-directions:

$$\mathbf{X}^\top\mathbf{X}\mathbf{q}_j = \mu_j\mathbf{q}_j.$$



Now rewrite the OLS solution in this basis. Define the data signal along direction $\mathbf{q}_j$ as

$$c_j = \mathbf{q}_j^\top\mathbf{X}^\top\mathbf{y}.$$

Then

$$\begin{aligned}\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} &= \mathbf{Q}\mathbf{\Lambda}^{-1}\mathbf{Q}^\top\mathbf{X}^\top\mathbf{y} \\ &= \sum_{j=1}^{d}\frac{c_j}{\mu_j}\mathbf{q}_j.\end{aligned}$$

Read one direction at a time

The OLS coordinate along $\mathbf{q}_j$ is

$$\boxed{\widehat{z}_j^{\mathrm{OLS}} = \frac{c_j}{\mu_j}.}$$

If μ_j is large, the direction is strongly supported by the predictors. If μ_j is tiny, even a small sample-dependent change in c_j is multiplied by the large factor $1/\mu_j$.



This is the exact mathematical bridge

Collinearity $\Rightarrow$ a direction with $\mathbf{X}\mathbf{q}_j \approx \mathbf{0}$ $\Rightarrow$ a small Gram eigenvalue $\mu_j = \|\mathbf{X}\mathbf{q}_j\|_2^2$ $\Rightarrow$ OLS amplification by $1/\mu_j$ $\Rightarrow$ high coefficient variance.

5. Ridge modifies the objective

Add a cost for large coefficients and derive the new solution

Ridge regression adds a squared ℓ_2 penalty:

$$\boxed{J_\lambda(\boldsymbol{\beta}) = \|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_2^2, \qquad \lambda \geq 0.}$$

fit the observed data $\|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2$ + **prefer smaller coefficients** $\lambda\|\boldsymbol{\beta}\|_2^2$

Step 1: differentiate both terms

$$\nabla_\beta J_\lambda(\boldsymbol{\beta}) = 2\mathbf{X}^\top(\mathbf{X}\boldsymbol{\beta} - \mathbf{y}) + 2\lambda\boldsymbol{\beta}.$$

Set the gradient to zero:

$$\mathbf{X}^\top\mathbf{X}\widehat{\boldsymbol{\beta}}_{\text{Ridge}} - \mathbf{X}^\top\mathbf{y} + \lambda\widehat{\boldsymbol{\beta}}_{\text{Ridge}} = \mathbf{0}.$$

Step 2: collect the coefficient vector

$$(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})\widehat{\boldsymbol{\beta}}_{\text{Ridge}} = \mathbf{X}^\top\mathbf{y},$$

$$\widehat{\boldsymbol{\beta}}_{\text{Ridge}} = \boxed{(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^\top\mathbf{y}}.$$

The only algebraic change from OLS is $\mathbf{X}^\top\mathbf{X} \mapsto \mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}$. Its consequences are substantial.

Step 3: see what happens to every eigenvalue

Because $\mathbf{I}\mathbf{q}_j = \mathbf{q}_j$,

$$\begin{aligned}(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})\mathbf{q}_j &= \mathbf{X}^\top\mathbf{X}\mathbf{q}_j + \lambda\mathbf{q}_j \\ &= \mu_j\mathbf{q}_j + \lambda\mathbf{q}_j \\ &= (\mu_j + \lambda)\mathbf{q}_j.\end{aligned}$$

Ridge leaves the eigen-directions unchanged and shifts every Gram eigenvalue upward by λ.

Therefore,

$$\begin{aligned}\widehat{\boldsymbol{\beta}}_{\text{Ridge}} &= \mathbf{Q}(\boldsymbol{\Lambda} + \lambda\mathbf{I})^{-1}\mathbf{Q}^\top\mathbf{X}^\top\mathbf{y} \\ &= \sum_{j=1}^{d} \frac{c_j}{\mu_j + \lambda}\mathbf{q}_j.\end{aligned}$$

Why the penalty stabilizes the inverse

OLS divides the signal c_j by μ_j. Ridge divides it by $\mu_j + \lambda$. A nearly zero denominator can no longer occur when $\lambda > 0$.

What Ridge does not do

Ridge does not delete the weak direction or change the observed data. It changes how aggressively the fitted model responds to evidence along that direction. In practice, predictors are standardized first and an intercept is usually left unpenalized.

6. Ridge shrinks direction by direction

The weak directions are changed the most

Recall

$$\hat{z}_j^{\mathrm{OLS}} = \frac{c_j}{\mu_j}, \qquad \hat{z}_j^{\mathrm{Ridge}} = \frac{c_j}{\mu_j + \lambda}.$$

Relate the two estimates:

$$\hat{z}_j^{\mathrm{Ridge}} = \underbrace{\frac{\mu_j}{\mu_j + \lambda}}_{s_j(\lambda) \in [0,1]} \hat{z}_j^{\mathrm{OLS}}.$$

The multiplier $s_j(\lambda)$ is the **direction-wise shrinkage factor**.

direction	Gram eigenvalue	λ	shrinkage $\mu_j/(\mu_j+\lambda)$	interpretation
strong	$\mu_1 = 100$	4	$100/104 \approx 0.962$	retained almost fully
weak	$\mu_2 = 1$	4	$1/5 = 0.20$	heavily shrunk



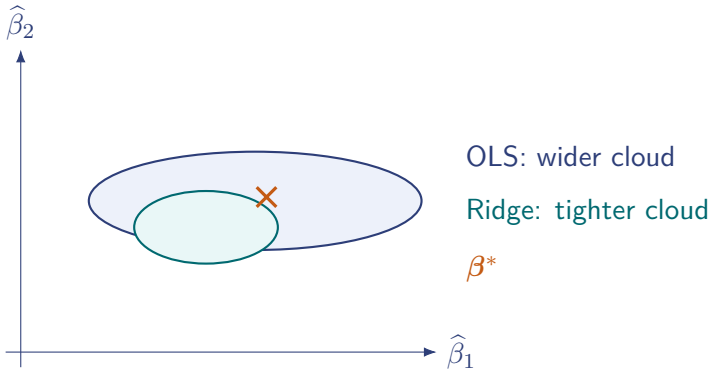
pale bar: OLS directional coordinate colored bar: Ridge coordinate

Why this is a bias–variance tradeoff

For a fixed dataset, Ridge moves the estimate away from OLS and toward zero. Across repeated datasets, however, the fitted coefficient cloud becomes tighter, especially along directions that the predictors barely support.

more λ $\Longrightarrow$ more shrinkage, more bias, usually less variance.

The best λ is therefore selected using validation data, not by minimizing the training error alone.



The Ridge cloud is tighter but its center is shifted toward zero.

Check your understanding

What happens to $s_j(\lambda)$ when (a) $\lambda = 0$, (b) $\lambda \to \infty$, and (c) μ_j is very large relative to λ?

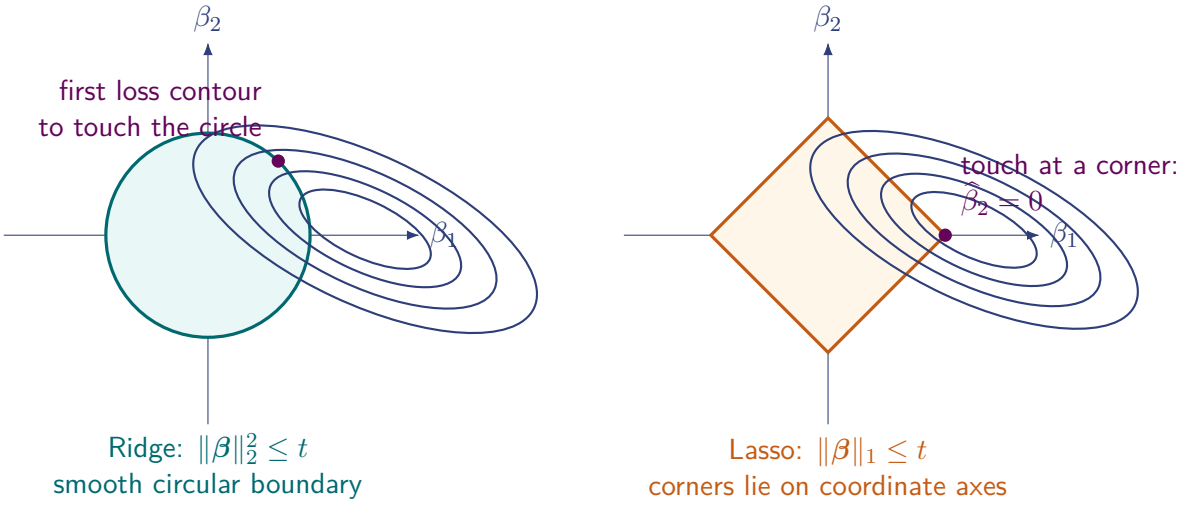
7. The constrained geometric view

A penalty and a coefficient budget describe the same tradeoff

Ridge can also be written as a constrained problem:

$$\min_{\boldsymbol{\beta}} \ \|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2 \quad \text{subject to} \quad \|\boldsymbol{\beta}\|_2^2 \le t.$$

The budget t limits how far the coefficient vector may move from the origin. For a corresponding choice of t, this constrained problem and the penalized problem have the same solution.



How the two Ridge formulations correspond

The Lagrangian of the constrained Ridge problem is

$$\mathcal{L}(\boldsymbol{\beta}, \lambda) = \|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2 + \lambda(\|\boldsymbol{\beta}\|_2^2 - t).$$

For optimization over $\boldsymbol{\beta}$, the constant $-\lambda t$ does not affect the minimizer, leaving the penalized Ridge objective. A smaller budget t corresponds to a stronger effective penalty, but in general $\lambda \neq 1/t$.

Why Lasso can perform feature selection

Replace the squared ℓ_2 penalty by an ℓ_1 penalty:

$$J_{\text{Lasso}}(\boldsymbol{\beta}) = \|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1, \qquad \|\boldsymbol{\beta}\|_1 = \sum_{j=1}^{d} |\beta_j|.$$

The diamond-shaped feasible region has corners on the coordinate axes. A loss contour can first touch at a corner, making one or more coefficients exactly zero. Ridge usually shrinks coefficients without setting them exactly to zero.

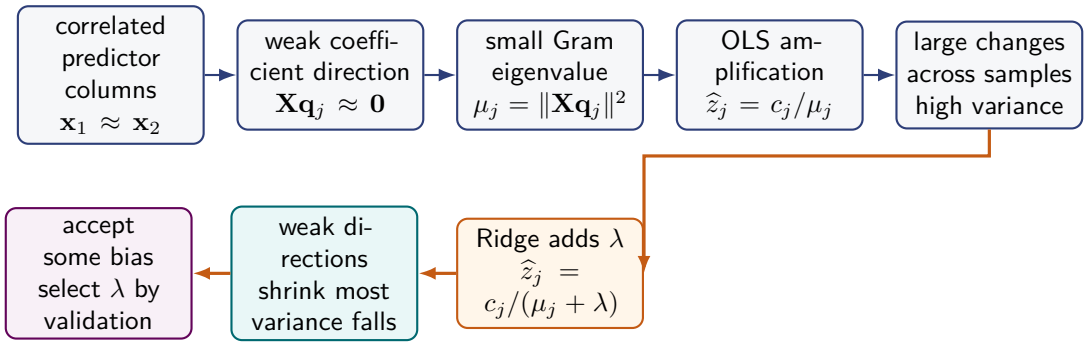
Elastic Net combines both preferences

$$J_{\text{EN}}(\boldsymbol{\beta}) = \|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2 + \lambda\Big(\alpha\|\boldsymbol{\beta}\|_1 + (1-\alpha)\|\boldsymbol{\beta}\|_2^2\Big), \quad 0 \le \alpha \le 1.$$

It mixes Lasso sparsity with Ridge stability, which is often useful when predictors are strongly correlated.

8. Put the entire lecture in one chain

What students should be able to explain without memorizing formulas



The answer to "why does this matter?"

The eigenvalues of $\mathbf{X}^\top\mathbf{X}$ tell us which combinations of coefficients the observed predictors can determine reliably. OLS trusts all of those directions equally after inversion. Ridge distrusts weak directions in a precise, graded way.

Method	Objective	Main effect
OLS	$\|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2$	best training fit; can be unstable in weak directions
Ridge	$\|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_2^2$	smooth direction-wise shrinkage; stabilizes correlated predictors
Lasso	$\|\mathbf{X}\boldsymbol{\beta} - \mathbf{y}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1$	shrinkage plus possible exact zeros; embedded feature selection
Elastic Net	squared loss plus ℓ_1 and ℓ_2 penalties	sparsity with additional stability for correlated predictors

Five questions that test understanding

1. Why can two very different coefficient vectors produce almost the same predictions when predictors are correlated?
2. Show that $\mu_j = \|\mathbf{X}\mathbf{q}_j\|_2^2$ for a unit eigenvector $\mathbf{q}_j$.
3. Where does the dangerous factor $1/\mu_j$ enter the OLS solution?
4. Why does Ridge shrink a weak direction more than a strong direction?
5. Why can Lasso produce an exact zero while Ridge usually does not?

Final distinction

Training fit asks how closely $\mathbf{X}\boldsymbol{\beta}$ matches the observed $\mathbf{y}$. **Stability** asks how much the fitted model changes when the training sample changes. Regularization may slightly worsen the first quantity in order to improve the second and, ultimately, prediction on unseen data.

Prepared from the classroom board notes. The mathematical sequence has been preserved, notation has been standardized, and explanatory steps have been added where the board discussion supplied the verbal bridge.